

NMAM INSTITUTE OF TECHNOLOGY, NITTE

(An Autonomous Institution affiliated to VTU, Belagavi)

Fifth Semester B.E. (Mechanical) (Credit System) Degree Examinations

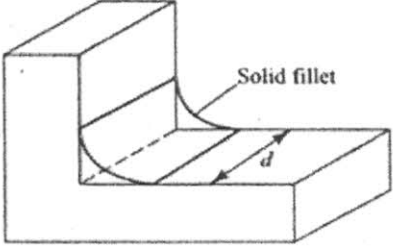
January 2023

20MEE51 / 17MEE51 – CAD / CAM

Duration: 3 Hours

Max. Marks: 100

Note: Answer **Five full questions** choosing **Two full questions** from **Unit – I & Unit – II each** and **One full question** from **Unit – III**.

Unit – I		Marks	BT*	CO*	PO*
1. a)	Define computer aided design (CAD) and Computer Aided Manufacturing (CAM). Explain how computers can be used in design process.	10	L*2	1	1
b)	Derive initial decision parameter (P_0) of Bresenham's line generation algorithm with suitable sketch.	10	L2	2	1
2. a)	Explain beam penetration and shadow mask technique to produce color display with suitable sketch.	10	L2	1	1
b)	Obtain the coordinates of pixels lying on the $(1/8)^{\text{th}}$ portion of a circle of radius 12 units centered on (0,0) and starting from (0, 12).	10	L3	2	1
3. a)	Rotate the square defined by corner points (2,2), (2,4), (4,4) and (4,2) about the corner (2,2) through 90 degree and obtain the new coordinates of the rotated square. Plot the rotated square.	6	L3	1	1
b)	Explain computer monitoring and control with a block diagram.	4	L2	1	1
c)	Obtain cubic spline equation with starting point (2,2) end point (14,14) with the start angle slope=30°, end angle slope = 30°. Also determine the points at $u=0, \frac{1}{4}, \frac{1}{2}, \frac{3}{4}, 1$	10	L3	2	2
Unit – II					
4. a)	Explain different types of synthetic surfaces with a suitable sketch.	10	L2	3	1
b)	Explain the different components of NC machine tool.	8	L2	4	1
c)	List the advantages of NC machine.	2	L2	4	1
5. a)	Define Half space and construct the solid using Half- spaces for the figure shown below.				
 (a) Solid fillet		10	L3	3	1
b)	Define CNC and explain different types of CNC.	10	L2	4	1
6. a)	With the suitable sketch, explain the meaning of different Euler operators used in Boundary representation of solids.	10	L2	3	1
b)	List different types of Adaptive control system and explain ACC system with a neat sketch.	10	L2	4	1
Unit – III					
7. a)	Define group technology. Explain how part families are generated. Give examples.	10	L2	5	1
b)	Explain the following: (i) Retrieval-type (ii) Generative CAPP systems.	10	L2	5	1
8. a)	Explain opitz coding system.	10	L2	5	1
b)	Explain shop floor control system and also discuss functions of shop floor control.	10	L2	5	1

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
